

ARTICLES

The efficient interaction of indirect reciprocity and costly punishment

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Human cooperation in social dilemmas challenges researchers from various disciplines. Here we combine advances in experimental economics and evolutionary biology that separately have shown that costly punishment and reputation formation, respectively, induce cooperation in social dilemmas. The mechanisms of punishment and reputation, however, substantially differ in their means for ‘disciplining’ non-cooperators. Direct punishment incurs salient costs for both the punisher and the punished, whereas reputation mechanisms discipline by withholding action, immediately saving costs for the ‘punisher’. Consequently, costly punishment may become extinct in environments in which effective reputation building—for example, through indirect reciprocity—provides a cheaper and powerful way to sustain cooperation. Unexpectedly, as we show here, punishment is maintained when a combination with reputation building is available, however, at a low level. Costly punishment acts are markedly reduced although not simply substituted by appreciating reputation. Indeed, the remaining punishment acts are concentrated on free-riders, who are most severely punished in the combination. When given a choice, subjects even prefer a combination of reputation building with costly punishment. The interaction between punishment and reputation building boosts cooperative efficiency. Because punishment and reputation building are omnipresent interacting forces in human societies, costly punishing should appear less destructive without losing its deterring force.

Human cooperation is a paradox. Although human altruism seems unique in the animal world¹ and we are no doubt champions of reciprocity², there is a kind of social dilemma³ that comes in many guises, some of which may eventually challenge our existence. We obviously overuse public resources—for example, by over-fishing oceans, driving pension and health insurance systems to bankruptcy, and risking the collapse of the global climate through unlimited use of fossil energy. These are showpieces of the “tragedy of the commons”, for which Hardin⁴ envisaged only inevitable breakdown.

However, several studies have found factors that can boost cooperation in social dilemmas, the most successful ones being costly punishment of defectors^{5–10} and reputation that is essential for success in interacting games^{11–13}. The ‘public goods game’ is the experimental model used to study social dilemmas¹⁴: for example, four players are given an endowment that they are free to keep for themselves or to contribute to a common pool. The content of the pool is then doubled and redistributed among the players irrespective of their contribution. If all contribute £1, everybody receives £2 (that is, a net gain of £1). If all players but one contribute, the defector has a net gain of £1.50 and the contributors £0.50 each. Individual self-interest results in defection and is at odds with group interest. Usually, initial cooperation declines quickly¹⁴, supporting Hardin’s views⁴. When players are allowed to punish others costly (that is, to reduce the others’ income at their own cost—for example, by investing £1 to reduce another player’s income by £3), defectors are heavily punished and typically increase their contributions in future rounds, even with new partners⁷. Such altruistic punishment leads to a large increase in cooperation, but induces another type of dilemma—namely, that cooperation enhancement through severe costly punishment considerably destroys monetary gains: both the punisher and the punished lose money. Especially in the early rounds, when

punishment is heavily used, efficiency (that is, realized payoff as a fraction of maximally attainable payoff) is lower than in the absence of the punishment option. Not until cooperation has been established on a stable level is a significantly higher efficiency achieved under altruistic punishment. Thus, the group would be better off if contributions to the public good could be induced by a less destructive means¹⁵.

Various alternative mechanisms, such as symbolic punishment¹⁶ and direct rewarding of other subjects¹⁷, have failed to solve the cooperation dilemma. However, if the public goods game is incorporated in a broader context that promises rewards for those with a good reputation—that is, an ‘indirect reciprocity game’—cooperation in the public goods game can be maintained at a high level¹¹. This reputation effect, ‘give and you shall receive’, is predicted to be rather robust¹². In indirect reciprocity situations^{2,18,19}, individuals who have helped others are given support—in other words, the supporters improve their reputation and are rewarded in turn^{20–23}. Players use withholding help in the indirect reciprocity game as a ‘punishment’ device for defectors in the public goods game. However, this device does not generate the dilemma of direct efficiency reduction of altruistic punishment: withholding help does not afford any monetary investment and the ‘punisher’ actually saves money by not rewarding a defector (who does not lose his money either; instead, he does not receive additional gains). The ‘punisher’ may not lose even in the long run, because an otherwise cooperative person will not destroy her image score by sometimes not giving in the indirect reciprocity game²⁴. Yet, ‘punishment’ through withholding help gives up the opportunity to increase efficiency through helping in the indirect reciprocity game. Thus, the accumulated evidence seems to suggest a straightforward prediction: in a social dilemma situation that allows for effective reputation building, costly punishment may

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become extinct because a much cheaper and equally powerful alternative to sustain cooperation is available. But if this is true, we face the following puzzle. Why is direct punishment of defectors a universal feature in all known human societies?

Here we try to solve this puzzle in an experiment with human volunteers. Groups of eight subjects each played 20 periods of the public goods game. We studied two treatments (Table 1). In treatment 'PUN&IR', before each period, each player can choose between joining a group in which the public goods game is followed by both costly punishing (PUN) and an indirect reciprocity (IR) game, and a group in which the public goods game is followed solely by an indirect reciprocity game; that is, we offer a choice between punishing plus indirect reciprocity and solely indirect reciprocity in treatment PUN&IR. In treatment 'PUN', before each period, each player can choose between joining a group in which the public goods game is followed by costly punishing and a group in which the public goods game is not combined with any other option; that is, we offer a choice between punishing and no additional option in treatment PUN⁹. Thus, players in PUN have the same choice as in PUN&IR but without an indirect reciprocity game in either option (see Fig. 1 and Supplementary Information). In each indirect reciprocity game, each player acts as both a 'sender' and a 'receiver'. The sender may help the receiver by sending money, which reaches the receiver in tripled amount. Before sending, the sender is informed about the receiver's 'reputation': the sender gets to know the contributions of the receiver in the actual period of the public goods game and in all previous periods, and about the previous indirect reciprocity sending decisions of the receiver. Direct reciprocity is excluded (see Methods).

Our experimental design studies a situation in which reputation cannot be avoided, but the punishment option can be chosen or avoided. We deliberately did not test whether people prefer a reputation option to a punishment option, both combined with a public goods game, because we think that this kind of choice is unlikely to occur in reality and thus humans may not be expected to have developed or evolved rules for such a decision. It seemed obvious to study whether a reputation-sensitive world can be invaded by a strategy of costly punishment because being watched can hardly be avoided in real life. We expected that players in PUN&IR would go directly for the punishment-free group because they would use the indirect reciprocity game for cooperation enhancement and avoid the detrimental effects of costly punishment^{9,15,25}. Below, we continue to use the terms 'punishment' or 'punisher' (in quotation marks) for withholding help in the indirect reciprocity game; the terms punishment or punisher are used for direct costly punishment (see refs 5–10).

The interaction is preferred and boosts efficiency

In both treatments, about 70% of the subjects initially chose the group without costly punishment (the punishment-free group or

PFG⁹). Contrary to expectations, however, subjects preferred the group with the opportunity of costly punishment (the punishment-opportunity group or POG) in the second half (periods 11–20) significantly in treatment PUN&IR ($N = 12$ groups, $z = -2.640$, $P = 0.008$, two-tailed Wilcoxon signed-ranks matched-pairs test) but not significantly in treatment PUN ($N = 6$ groups, $z = -1.787$, $P = 0.074$).

The contributions to the public goods pool were significantly higher in the POG than in the PFG both in PUN&IR ($z = -3.061$, $P = 0.002$, two-tailed Wilcoxon signed-ranks matched-pairs test) and in PUN ($z = -2.226$, $P = 0.026$; Fig. 2a). However, the contributions, either 0 or 10 money units (MUs), in the POG became significantly higher in PUN&IR than in PUN in the second half ($z = -1.976$, $P = 0.048$, two-tailed Mann–Whitney U -test). This result was achieved because in the POG the frequency of contributions of 10 MUs was higher in PUN&IR than in PUN (Fig. 2b, c; $z = -2.062$, $P = 0.039$).

The interaction reduces but concentrates punishment

The interaction between costly punishment and indirect reciprocity not only increased the contributions to the public good but also significantly reduced the number of punishment points allocated per member of the POG in PUN&IR as compared with PUN (Fig. 3a; $z = -2.767$, $P = 0.006$, two-tailed Mann–Whitney U -test). Nevertheless, the punished free-riders were more heavily punished in PUN&IR than in PUN (Fig. 3b; $z = -2.313$, $P = 0.018$). Thus, although fewer group members were punished, punishment of a free-rider, that is a non-cooperator, was even more severe in the presence of the indirect reciprocity game than in its absence. This

Table 1 | Experimental design

Endogenous choice	Experimental treatments (exogenous)			
	PUN&IR		PUN	
	Indirect reciprocity exists (reputation building possible)		Indirect reciprocity ruled out (reputation building impossible)	
POG Public good with punishment opportunity	PUN&IR	POG	PUN	POG
PFG Public good without punishment opportunity	PUN&IR	PFG	PUN	PFG

PFG, punishment free group; POG, punishment opportunity group; PUN, public good game with costly punishment opportunity; PUN&IR, public good game with costly punishment opportunity and indirect reciprocity game.

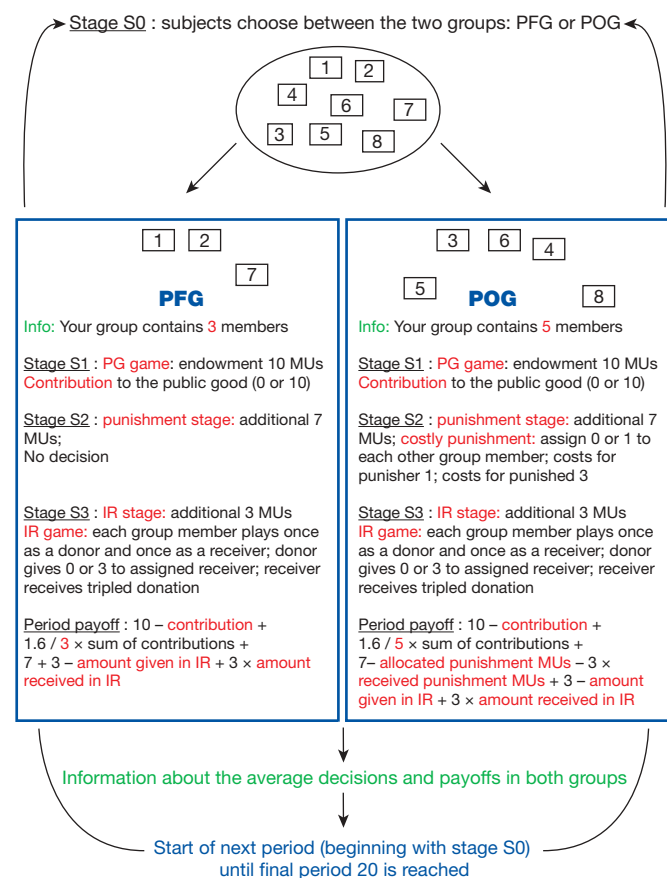


Figure 1 | The course of a period. Shown is the course of a period in treatment PUN&IR with the stages S0, S1, S2 and S3. In treatment PUN, the stages S0, S1 and S2 are played as shown and the period ends after the termination of stage S2. In all treatments, each of the 20 periods has an identical structure. PG game, public goods game.

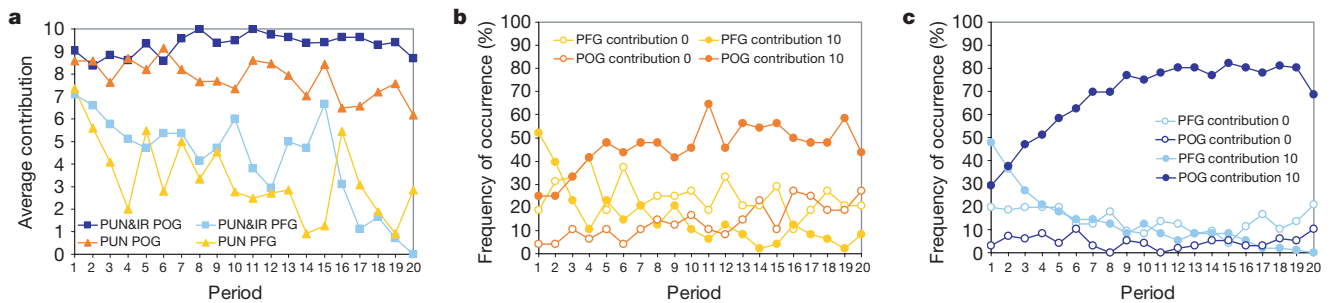


Figure 2 | Contributions to the public good. **a**, Contributions to the public good are higher in the POG than in the PFG in both treatments. **b**, In treatment PUN, subjects contributing 10 MUs become a relative, but never

is notable, because it might have been expected that the costly punishment acts would be avoided completely at this stage and that punishment would be moved (denoted 'punishment') to the indirect reciprocity stage. Contributions to the public goods pool per allocated punishing point are about three times higher in PUN&IR than in PUN (Fig. 3c; $z = -2.363$, $P = 0.015$). Thus, subjects could generate higher contributions with fewer punishment expenses (Fig. 3d; $z = -2.160$, $P = 0.030$ in the second half).

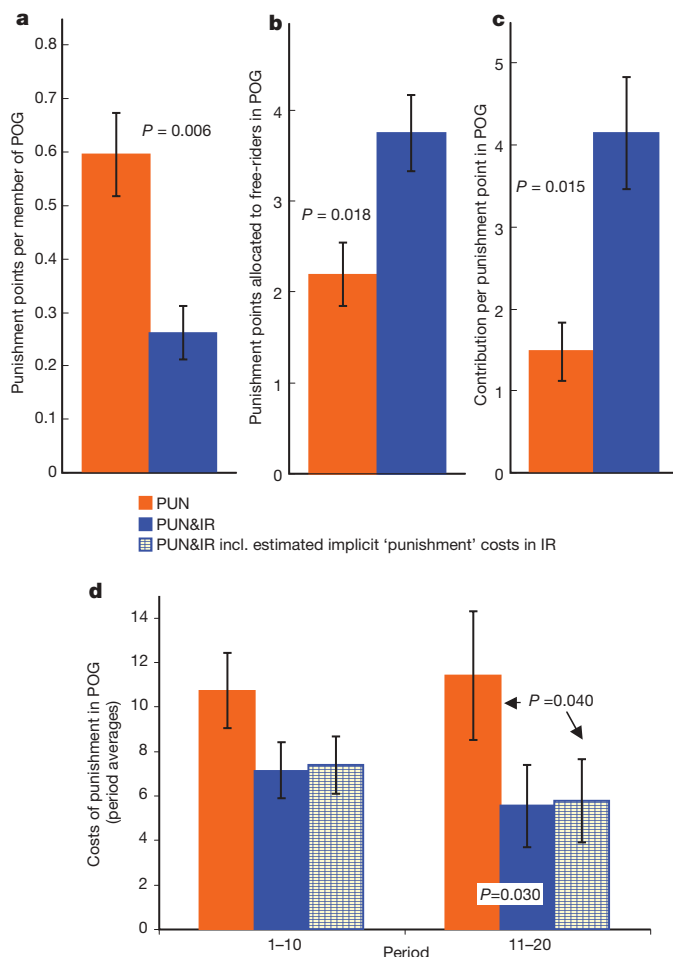


Figure 3 | Punishment behaviour. **a**, The punishment points allocated per member are significantly higher in PUN than in PUN&IR. **b**, Free-riders are significantly more severely punished in PUN&IR than in PUN. **c**, The contribution achieved per punishment point is significantly higher in PUN&IR than in PUN. **d**, The punishment costs are higher in PUN than in PUN&IR, even if the estimated implicit costs of 'punishment' in the indirect reciprocity stage are added, especially in the second half (see text for statistics). Data are the mean $\pm$ s.e.m. per group.

an absolute majority. **c**, In treatment PUN&IR, subjects contributing 10 MUs quickly become an absolute majority. Data are the mean per group for each period.

In addition, when the costs of punishment of both the punisher and the punished were added to the profits from the public goods game, those who contributed 10 MUs to the public pool in the POG came very close to the social optimum—that is, full contribution of all subjects and no punishment acts (Fig. 4a, horizontal line at 27)—and had a higher net payoff in PUN&IR than in PUN ($z = -2.435$, $P = 0.015$; two-tailed Mann–Whitney U -test). Those who contributed 0 MUs were not significantly better off in PUN&IR than in PUN ($z = -1.032$, $P = 0.302$). In both treatments, their payoff seems to come close to the Nash equilibrium of all players defecting and no punishment acts executed (Fig. 4a, horizontal line at 17). Even if we compare the efficiency in the public goods game including all costs of punishment of the combination of both options (with and without punishment) of a treatment, we find that PUN&IR is more efficient

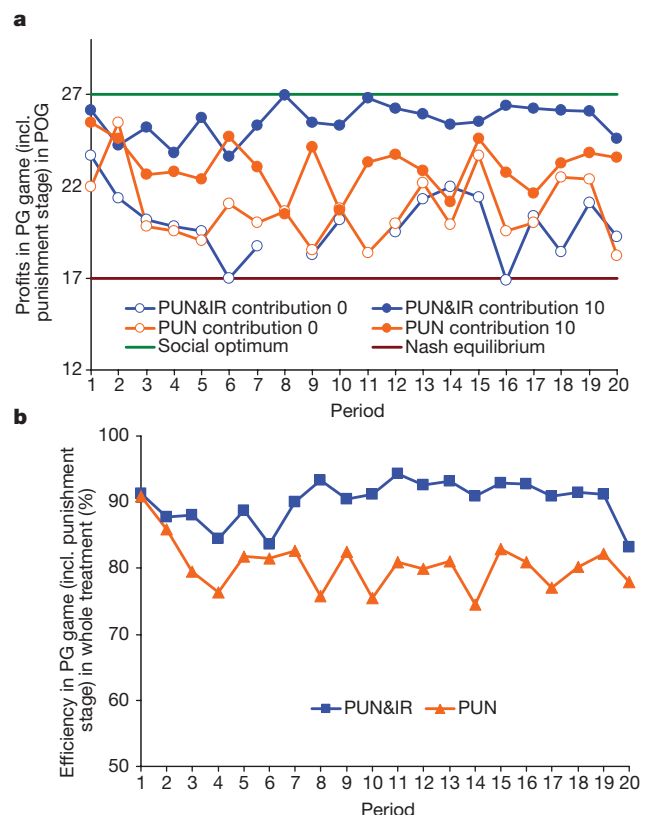


Figure 4 | Payoffs in the public goods game including the punishment stage and efficiency. **a**, Payoffs are highest for the full contributors in the POG in PUN&IR. Horizontal line at 27 indicates the social optimum; horizontal line at 17 indicates the Nash equilibrium. **b**, Treatment efficiency, that is, the efficiency of the whole treatment combining both groups (PFG and POG), is higher in PUN&IR than in PUN. Data are the mean per group.

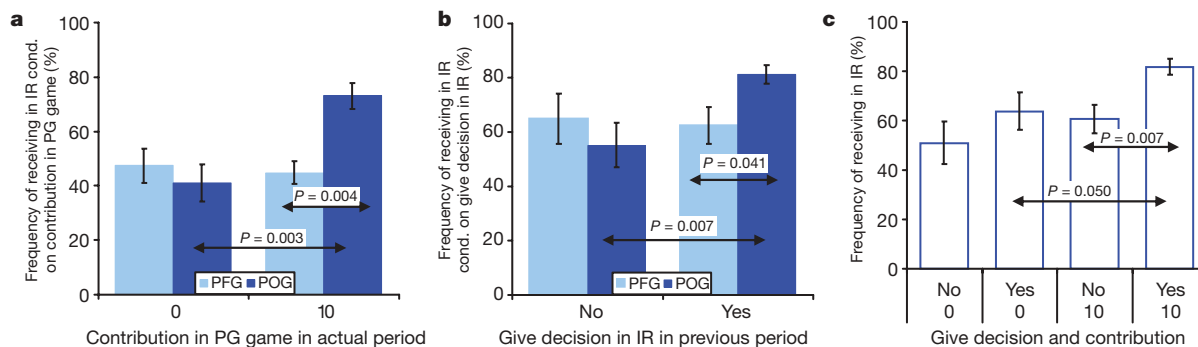


Figure 5 | Receiving in the indirect reciprocity game. **a**, High contributors in the public goods game receive significantly more often in the indirect reciprocity (IR) game in the POG. **b**, Subjects who gave in the previous period of the indirect reciprocity game receive significantly more often in the actual period than subjects who did not give in the previous period. **c**, High

contributors in the public goods game who gave in the previous period of the indirect reciprocity game receive significantly more often than high contributors in the public goods game who did not give in the previous period of the indirect reciprocity game. Data are the mean $\pm$ s.e.m. per group (see text for statistics).

than PUN (Fig. 4b; $z = -2.557$, $P = 0.010$, two-tailed Mann–Whitney U -test).

These results suggest that the presence of the indirect reciprocity game has a positive influence on the cooperation behaviour in the public goods game. Indeed, we find an interaction among indirect reciprocity, costly punishment and contributions to the public pool. Those subjects who had contributed 10 MUs to the public pool received significantly more often in the indirect reciprocity game than those who had contributed nothing (Fig. 5a; $z = -2.934$, $P = 0.003$, two-tailed Wilcoxon signed-ranks matched-pairs test) in the POG but not in the PFG ($z = -0.236$, $P = 0.814$). In addition, contributing 10 MUs was more often ‘rewarded’ in the POG than in the PFG ($z = -2.845$, $P = 0.004$). Thus, defectors in the POG are hit twice: they are punished once in the punishment stage and ‘punished’ a second time in the indirect reciprocity game.

Implicit costs of ‘punishment’ in the indirect reciprocity game

In addition to the calculated explicit costs of punishment in the punishment stage, the ‘punishment’ in the indirect reciprocity game induces implicit ‘punishment’ costs by giving up the opportunity to increase the efficiency by helping. The implicit ‘punishment’ costs can be estimated from the difference in the likelihood of receiving between subjects who contributed to the public good and those who did not, under the condition that they had the same giving behaviour in the previous period of the indirect reciprocity game. For example, a contributor who gave in the previous period of the indirect reciprocity game receives with a likelihood of 85.3%, whereas a free-rider who gave in the previous period of the indirect reciprocity game receives with a likelihood of 59.1%. The assumption that the difference in the likelihood of receiving between a contributor and a free-rider is attributed solely to the ‘punishment’ of free-riders in the indirect reciprocity game provides an upper bound for the implicit ‘punishment’ costs in the indirect reciprocity game. The estimation of these implicit ‘punishment’ costs shows that they are extremely low (on average, 0.30 per period in the first ten periods and 0.23 per period in the last ten periods). Thus, adding these implicit ‘punishment’ costs to the explicit punishment costs (Fig. 3d) does not change the picture that the interaction of costly punishment and reputation building through indirect reciprocity increases efficiency (the punishment costs in PUN are higher than the sum of the explicit punishment and the implicit ‘punishment’ costs in PUN&IR: $z = -2.064$, $P = 0.040$).

Reputation building is effective in the interaction

The indirect reciprocity game ‘worked’ by itself only in the POG. Individuals who had helped others in the previous indirect reciprocity game were significantly more likely to receive money in the current indirect reciprocity game (Fig. 5b) only in the POG

($z = -2.714$, $P = 0.007$, in PFG: $z = -0.561$, $P = 0.574$, two-tailed Wilcoxon signed-ranks matched-pairs test). Thus, giving in the indirect reciprocity game was more often rewarded in the POG than in the PFG ($z = -2.045$, $P = 0.041$). The efficiency (that is, the percentage of the maximal attainable payoff actually achieved) of indirect reciprocity was significantly higher in the POG than in the PFG ($z = -2.805$, $P = 0.005$). These results suggest that the subjects who chose the POG cared more about other people’s reputation than those who avoided the POG. It would be interesting to study the motivation of the few subjects who finally decided to avoid the POG in PUN&IR. These subjects seem to be pure egoists who prefer not to have a direct punishment opportunity and do not discriminate in their helping decisions between those with a low or a high reputation or between those who cooperated and those who defected in the public goods game. Nevertheless, in both groups the ‘good guys’ in the public goods game were likely to be ‘donors’ in the indirect reciprocity game (POG: $\rho = 0.738$, $P = 0.01$; PFG: $\rho = 0.887$, $P = 0.0001$, Spearman rank correlation).

Those who contributed to the public pool thus seem more at risk to damage their reputation by not giving in indirect reciprocity rounds than those who defected in the public goods game. Giving or not giving in the indirect reciprocity game was significantly taken into account in the next indirect reciprocity game when both kinds of receivers had contributed 10 MUs to the public pool (Fig. 5c; $z = -2.707$, $P = 0.007$, two-tailed Wilcoxon signed-ranks matched-pairs test) but not when receivers had not contributed ($z = -1.472$, $P = 0.141$).

Free-riders are disciplined more in the interaction

We indicated above that the ‘double punishment’ in PUN&IR increased the costs of defecting in the public goods game. To study the consequences for individual subjects, we investigated the behaviour of the two least cooperative subjects of each group, that is, those who contributed least and second least to the public goods pool during the 20 periods. On average, these two defected in 7.4 ± 1.2 periods (mean $\pm$ s.e.m.) in PUN&IR and in 12.3 ± 1.9 periods in PUN ($z = -2.255$, $P = 0.024$, two-tailed Mann–Whitney U -test). Notably, the number of periods in which the two least cooperative subjects chose the POG was not significantly different between PUN&IR (12.8 ± 1.5 times) and PUN (11.0 ± 1.6 times, $z = -1.176$, $P = 0.240$). However, they defected significantly less often when in the POG in PUN&IR (2.6 ± 0.8 times) than in the POG in PUN (5.3 ± 0.7 times, $z = -2.674$, $P = 0.007$). Thus, the least cooperative subjects were more cooperative in PUN&IR than in PUN, and they did not avoid the POG in either treatment.

Discussion and general implications

The interaction between the potential for costly punishment and building personal reputation in public goods games produced

unexpected results. Although subjects could choose a punishment-free option where reputation building alone would allow for high levels of cooperation in a public goods game^{11,12}, the great majority chose the punishment option that was also combined with reputation building. It is worthwhile to mention that most subjects preferred the group with the punishment option even though the private benefits from one unit contribution to the public good are diluted by the larger number of group members. However, the interaction with indirect reciprocity reduced punishment behaviour to almost a third and thus largely removed the efficiency dilemma of costly punishment. Indirect reciprocity did not simply replace punishment. We observed not only a net increase in contributions to the public goods pool as compared with a control (PUN), but also a boost in efficiency. This marked effect was achieved by concentrating the remaining punishing acts on the smaller number of free-riders in PUN&IR, which were thus more heavily punished than the free-riders in PUN. Moreover, the few free-riders in PUN&IR were hurt a second time through withheld help in the indirect reciprocity game, which amplified the punishment effect. These strong incentives to cooperate probably induced not only the majority of subjects but even the 'worst' free-riders per group to be more cooperative when punishment was combined with reputation building.

No theoretical prediction from a mathematical model capturing the specific situation of our study has been available; our unexpected results might thus be an incentive for theorists to study the interaction between punishment and reputation building formally. Such an analysis may also provide deeper insights into the sensitivity of the interaction between punishment and reputation building concerning, for example, group size, group composition, completeness of information about others, or impact of punishment power.

Punishment seems to have a solid foundation in human interactions. Using and reacting to costly punishment has an emotional representation in the brain^{26,27} and thus seems to be a 'hardwired' prerequisite of our social life. In addition, the finding that we take sophisticated signs of 'others may watch' into account^{28,29} suggests that 'don't lose your reputation'³⁰ is an unconscious imperative that guides social behaviour. Punishment and reputation do not seem to be substitutes that may easily replace one another, but omnipresent interacting mechanisms. Costly punishment can evolve because it purchases the reputation of someone who demands cooperation in the public goods game³¹. Of note, overwhelmingly social and fair outcomes are achieved in a model when punishment and reputation in groups of three players in spatial public goods games on a lattice are combined³². Reputation mechanisms generate an environment in which the execution of punishment has to be less frequent, however, without taking away its deterring force. The interaction of both mechanisms not only comes closer to real life but also provides a convenient, low-cost solution to social dilemmas, keeping the teeth of costly punishment covered but ready to bite. The attenuation of the destructive force of punishment through reputation may help to explain its evolutionary survival. The study of the interaction of two obviously unavoidable mechanisms, punishment and reputation, for maintaining human cooperation will certainly help to disentangle more of our complex social and unsocial nature in the future.

METHODS

Subjects. A total of 144 undergraduate students from the University of Erfurt voluntarily participated in 18 experimental sessions with eight subjects each.

Game. Each of the 20 experimental periods (Fig. 1) proceeded with a group choice stage (S0), a voluntary contribution stage (S1), a punishment opportunity stage (S2), and an indirect reciprocity stage (S3). In treatment PUN&IR, all four stages were played in 12 independent sessions; in treatment PUN, the stages S0, S1 and S2 were played in 6 independent sessions.

In stage S0, the subjects simultaneously and independently choose between a POG and a PFG in which punishment is not possible. Each subject is informed about the number of subjects in each group. The remaining period interactions take place solely within the group chosen in S0. In stage S1, the public goods game, each subject is endowed with 10 MUs and may contribute either 0 or

10 MUs. The contributions are multiplied by 1.6 to create a benefit for the whole group that is equally distributed among the group members; in other words, if a group consists of n members, each member profits by $1.6/n$ MUs from each MU contributed. The MUs not contributed are transferred to the subject's private account.

At the beginning of stage S2, each subject receives an additional 7 MUs independent of the affiliation choice in S0. In the PFG, these MUs are directly transferred to the subject's private account without any decisions required. In the POG, these MUs may be used to punish other members of the POG. Punishing costs the punisher 1 MU and the punished subject 3 MUs. The MUs not used for punishment are transferred to the subject's private account.

In stage S3, each subject receives an additional 3 MUs independent of the affiliation choice in S0. These MUs may be sent to another member of the same group, the 'receiver'. The MUs are tripled when sent, such that the receiver achieves 9 MUs. If the sender does not send, the sender keeps the 3 MUs and nothing reaches the receiver. The sender is informed both about the contributions of the receiver in the actual period of the public goods game and in all previous periods, and about the previous indirect reciprocity sending decisions of the receiver in addition to the contributions to the public good of the past receiver's receiver. Each subject acts as both a sender and a receiver in each period. To exclude direct reciprocity, the matching between the sender and the receiver ensures that the receiver has not previously been the sender's sender and will not be in the future.

At the end of the period each participant receives anonymous summary information about both groups. At the beginning of the experiment, subjects received written and oral instructions. At the end of the experiment, subjects privately received their experimental earnings in cash. All experimental decisions were made on a computer screen using the experimental software Z-Tree³³.

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Supplementary Information is linked to the online version of the paper at www.nature.com/nature.

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